

Python Full Stack Web Development – Task 1

User Management Web Application

Project Description

This project is a User Management Web Application developed using Python Flask, HTML, CSS, and SQLite. The application allows users to enter their details through a web form. The submitted details are stored in an SQLite database and displayed in a user table.

Technologies Used

- Python
- Flask
- HTML
- CSS
- SQLite
- VS Code
- Web Browser

Project Structure

```
python-fullstack-task1/  
■  
■■■ app.py  
■■■ database.db  
■■■ README.md  
■  
■■■ templates/  
■   ■■■ index.html  
■  
■■■ static/  
    ■■■ style.css
```

Features

- User registration form
- Add user name and email
- Store user details in SQLite database
- Display all registered users
- Simple HTML/CSS interface
- Flask backend for handling requests
- SQLite database for storing user information

How the Application Works

1. The user opens the Flask web application.
2. The home page displays a form for entering user details.
3. The user enters their name and email.
4. The form sends the information to the Flask backend.
5. Flask receives the submitted data.
6. The data is stored in the SQLite database.
7. The application retrieves the stored users.
8. The users are displayed in a table on the web page.

Database

The project uses SQLite for data storage. The database file is **database.db**.

Users table:

id – Unique user ID

name – User name

email – User email

How to Run the Project

1. Open the project folder:

```
cd C:\python-fullstack-task1
```

2. Start the Flask application:

```
py app.py
```

3. Open the application in a web browser:

```
http://127.0.0.1:5000
```

Requirements

Python and Flask should be installed. Check Python with **py --version** and Flask with **py -m flask --version**.

Project Objective

The objective of this task is to build a basic User Management Web Application using Python Flask, HTML, CSS, and SQLite, including a form for adding users, backend processing, database storage, and displaying stored users.

Author

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